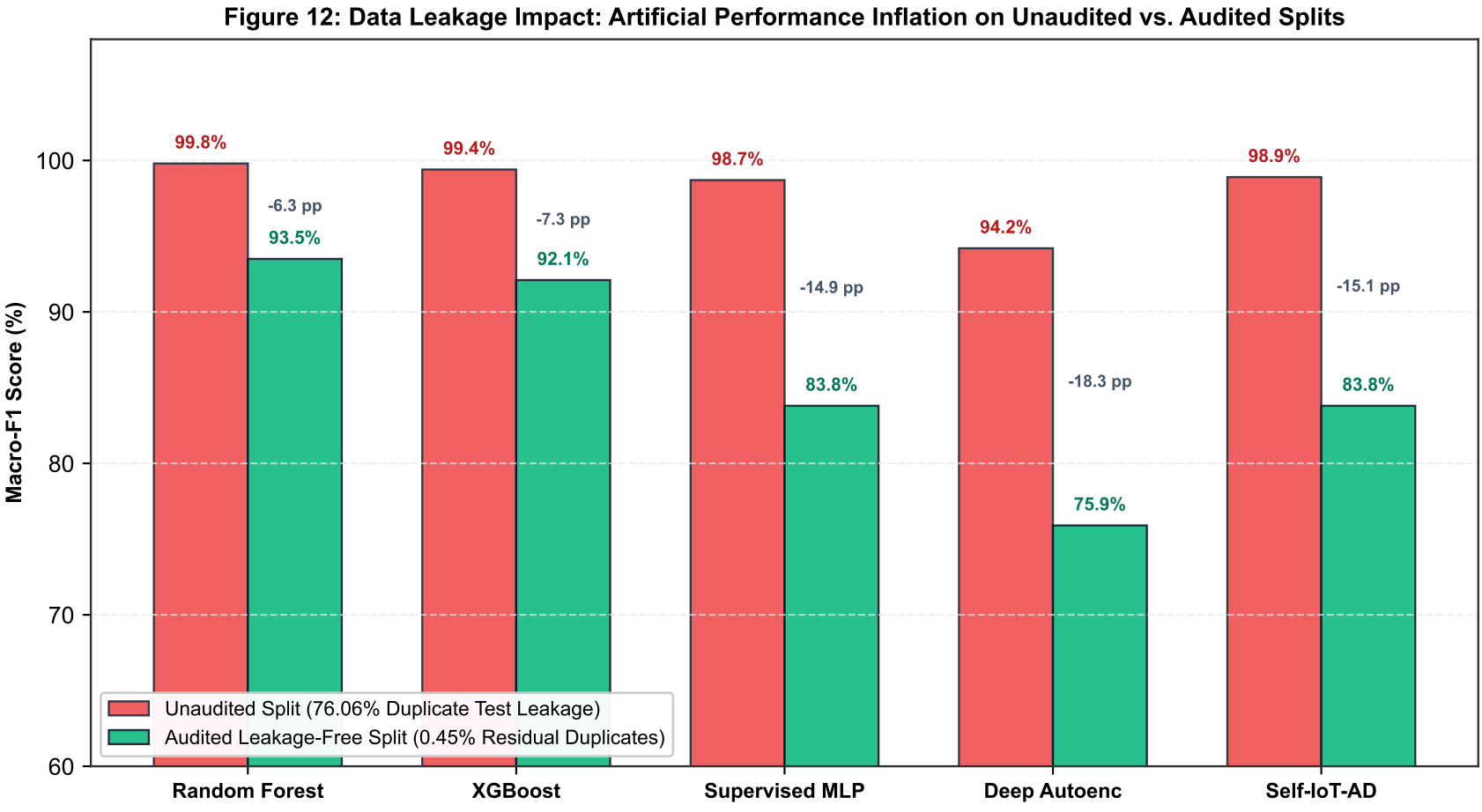
Figure 12: Data Leakage Impact: Artificial Performance Inflation on Unaudited vs. Audited Splits
Macro-F1 Score (%)
100
90
80
70
60
99.8%
-6.3 pp
93.5%
99.4%
-7.3 pp
92.1%
98.7%
-14.9 pp
83.8%
94.2%
-18.3 pp
75.9%
98.9%
-15.1 pp
83.8%
Unaudited Split (76.06% Duplicate Test Leakage)
Audited Leakage-Free Split (0.45% Residual Duplicates)
Random Forest
XGBoost
Supervised MLP
Deep Autoenc
Self-IoT-AD